

From Primitives to Photonic Chern Channels and Quantized Hall Drift

A Walkthrough of the Event Density Derivation (with Appendix A: Chern Integer-Quantization)

Allen Proxmire

May 2026

1 The Question

What this walkthrough derives

This walkthrough derives, from substrate primitives, the structural backbone of photonic Chern insulators:

1. **Synthetic frequency lattice** as a periodic rule-type substrate, with frequency modes as participation-rule indices.
2. **Haldane-type effective Hamiltonian** with nearest-neighbor (NN) + complex-phase next-nearest-neighbor (NNN) couplings on a honeycomb-like rule-type lattice.
3. **Berry curvature on the synthetic Brillouin zone**, concentrated near synthetic Dirac points and integrating to a non-zero quantity.
4. **Chern number** $C_n = (1/2\pi) \iint_{\text{BZ}} \Omega_n d^2k$ as the integrated global rule-type curvature, with integer-quantization derived in Appendix A.
5. **Quantized Hall drift**: the substrate-level transverse displacement of a minimal channel (light) under imposed ED-tension (synthetic electric field) is

$$\Delta x_{\perp} \text{ per cycle} = C_n \cdot a,$$

where a is the lattice period and C_n is the integer-valued Chern number of the occupied band.

6. **Driven-dissipative steady-state pinning**: the quantization survives Lindblad-type driven-dissipative dynamics because dissipation suppresses local fluctuations but cannot alter the global rule-type curvature.

The walkthrough is fully self-contained: every required Hilbert-space, eigenvalue, Bloch-form, gauge-transformation, semiclassical-equation-of-motion, and Lindblad step appears inside the document. The Chern-quantization proof is in Appendix A.

What standard photonics says, and where it stops

The Chénier et al. 2026 PRX experiment encodes a Haldane-like model in the synthetic frequency dimension of an optical fiber loop. Electro-optic modulation creates tunable NN and NNN couplings between frequency modes; complex modulation phases break time-reversal symmetry. The resulting synthetic lattice exhibits a honeycomb-like band structure, tunable topological phase transitions at non-zero Chern number, measurable Berry curvature across the Brillouin zone, and a quantized transverse drift of light under detuning of the modulation frequency.

The standard description treats this as a photonic analog of the integer quantum Hall effect: synthetic gauge fields produce non-trivial band topology, and the topology produces a quantized response. This is mathematically correct. It is also mechanistically opaque about *why* light,

despite carrying no charge, undergoes a quantized response; *what* the synthetic gauge field is at any deeper ontological level; and *why* the quantization is robust against driven-dissipative dynamics.

What Event Density claims

The quantized Hall drift is FORCED at the substrate level of the Event Density (ED) framework whenever a minimal ED-channel (light) propagates through a periodic rule-type substrate with non-zero global rule-type curvature. The mechanism is structural:

- Light is a minimal ED-channel that carries no internal multiplicity capable of imposing geometry; it must follow the rule-type structure of the substrate it propagates through.
- A periodic rule-type substrate forces Bloch-form eigenchannels and a band structure indexed by quasi-momentum on a Brillouin zone.
- Complex NNN couplings impose a global rule-type twist (Berry curvature) that cannot be removed by local rule rephasing.
- The integrated curvature is integer-valued (Chern number) by topology of the parameter-space bundle (Appendix A).
- Imposed ED-tension (synthetic electric field) forces the chain to slide along the curved manifold, producing a transverse displacement quantized in units of $C_n \cdot a$ per cycle.
- Driven-dissipative dynamics suppress local fluctuations but cannot deform the global topological invariant; the quantization survives.

Light does not need charge. It needs a periodic rule-type substrate with non-zero Chern number — and an imposed ED-tension to make the global curvature manifest as observable transport.

The substrate-level mechanism, in one sentence: **quantized Hall drift is the rule-type curvature integrated over a closed Brillouin zone, made observable as transport by ED-tension acting on a minimal channel.**

The chain in summary

The derivation chain runs: substrate primitives (§2) → synthetic frequency lattice as periodic rule-type substrate, with translation operator and Bloch form derived inline (§3) → Haldane-type effective Hamiltonian + Berry connection + Berry curvature across the BZ derived inline (§4) → Chern number as integrated curvature, integer-quantization derived in Appendix A (§5) → semiclassical equations of motion + quantized Hall drift (§6) → driven-dissipative Lindblad steady state with topological invariant preserved (§7) → substrate-level reading (§8) → forced/inherited/open accounting (§9) → exact claims (§10) → integer-quantization proof (Appendix A).

2 The Primitives

2.1 P-PC-1. Minimal ED-channel (photon)

A *photon* is a minimal ED-channel: a participation pathway with multiplicity $M = 1$. At any one substrate-tick, the channel commits to exactly one alignment thread. The minimal channel cannot store internal geometry — it has no internal multiplicity to deform — and therefore cannot impose structure on its substrate. It can only follow the rule-type structure of the surrounding substrate.

Substrate-level meaning. The photon is a probe: it reveals the substrate’s rule-type geometry as transport. Any curvature, twist, or tension in the substrate is expressed directly in the photon’s pre-individuation amplitude.

No charge requirement. The minimal-channel structure makes no reference to electric charge. The photon couples to the substrate’s rule-type structure regardless of whether that structure originates in a magnetic field, a synthetic gauge field, or a periodic dielectric pattern.

2.2 P-PC-2. Periodic participation rule in synthetic dimension

A *synthetic dimension* is a re-indexing of participation rules: distinct participation rules are labeled by an integer index $m \in \mathbb{Z}$, with the rule structure and inter-rule couplings determined by the substrate’s effective construction.

Periodicity. A periodic rule structure in synthetic dimension means

$$r_m = r_{m+N_{\text{period}}} \quad \text{or} \quad r_{m+a} = r_m$$

where a is the unit-cell size in synthetic-index units.

Frequency-mode realization. In the Chénier et al. experiment, frequency modes $\omega_m = \omega_0 + m\Omega_R$ (spaced by the cavity round-trip frequency Ω_R) play the role of synthetic-dimension sites. Electro-optic modulation imposes structured couplings between these modes. Frequency modes are not physical lattice sites; they are distinct timing alignments of ED-flow, indexed by m , that the substrate’s modulation pattern couples in a structured way.

2.3 P-PC-3. Complex rule-type couplings (NN + NNN)

Two types of rule-type couplings appear in the Haldane construction:

- **Nearest-neighbor (NN)** couplings: matrix elements of the effective Hamiltonian connecting mode m to mode $m + 1$, real-valued, denoted t_1 .
- **Next-nearest-neighbor (NNN)** couplings: matrix elements connecting mode m to mode $m + 2$, complex-valued with phase ϕ , denoted $t_2 e^{i\phi}$.

Substrate-level meaning. NN couplings are local ED-flow continuity: they encode how the chain’s identity transitions between adjacent participation rules. NNN couplings with complex phase are *global ED-twist*: a phase factor that cannot be removed by local rephasing of individual rules, because the phase accumulates around closed loops in the lattice.

Why complex. A real coupling produces no phase accumulation around any loop; a complex coupling with a phase ϕ produces a phase $e^{i\phi}$ for each NNN hop. When the lattice geometry contains closed paths (e.g., the honeycomb plaquettes of the Haldane model), the accumulated phase per plaquette is the substrate-level rule-type *flux* through that plaquette.

2.4 P-PC-4. Time-reversal breaking as directional ED-bias

Time-reversal symmetry of the substrate’s rule-type structure is the property that the rule structure is invariant under reversal of substrate-tick direction. Algebraically, the effective Hamiltonian satisfies $\mathcal{T}H\mathcal{T}^{-1} = H$ where $\mathcal{T}$ is the antiunitary time-reversal operator.

Breaking via complex NNN phase. The complex NNN coupling $t_2 e^{i\phi}$ breaks time-reversal symmetry when $\phi \neq 0, \pi$: the antiunitary $\mathcal{T}$ would map $t_2 e^{i\phi} \rightarrow t_2 e^{-i\phi}$, and these are distinct unless ϕ is a multiple of π .

Substrate-level meaning. Breaking time-reversal symmetry imposes a *directional bias* in the substrate’s rule-type structure: the rule structure distinguishes “forward” from “backward” propagation around closed loops. This directional bias is the substrate-level origin of the global ED-twist quantified by non-zero Chern number.

2.5 P-PC-5. ED-tension (synthetic electric field)

An *ED-tension* in synthetic dimension is a linear gradient in rule-type energy: the on-site energy of mode m takes the form

$$E_m = m \cdot F \cdot a$$

where F is the ED-tension strength and a is the lattice period. The total Hamiltonian becomes $H = H_{\text{lattice}} + F\hat{x}$, where $\hat{x}$ is the position operator on the synthetic lattice ($\hat{x}|m\rangle = ma|m\rangle$).

Realization. In the Chénier et al. experiment, detuning the modulation frequency by δ produces a linear shift in mode energies: $E_m = m\delta$. The ED-tension is the substrate-level analog of a synthetic electric field acting on the chain’s pre-individuation amplitude.

Substrate-level meaning. ED-tension is a structural tilt of the rule-type energy landscape that biases the chain’s participation flow. The chain seeks to lower its rule-type energy, which corresponds to drift along the synthetic dimension. When the substrate has zero rule-type curvature, this drift is purely longitudinal. When the substrate has non-zero rule-type curvature, the drift acquires a transverse component — the Hall drift.

2.6 P-PC-6. Driven-dissipative ED-flow equilibrium

Open photonic systems involve continual injection of substrate participation (the drive) and extraction (dissipation, including loss to bath modes and intentional decay channels). The substrate-level state is not a closed-system pure state but a steady-state distribution over alignment threads.

Lindblad-form coarse-graining. The substrate-level steady-state density operator ρ satisfies

$$\partial_t \rho = -i[H, \rho] + \sum_{\alpha} \left[L_{\alpha} \rho L_{\alpha}^{\dagger} - \frac{1}{2} \{ L_{\alpha}^{\dagger} L_{\alpha}, \rho \} \right] \quad (\partial_t \rho = 0 \text{ for steady state})$$

where H is the unitary-evolution generator, L_{α} are the Lindblad jump operators encoding the coupling to the bath (drive + loss). Derivation of this form is given inline in §7.

Steady state. The steady state is the unique stationary solution ρ_{ss} with $\partial_t \rho_{\text{ss}} = 0$. Under generic Lindblad operators, the steady state exists, is unique, and is reached on a timescale set by the dissipation rate.

Substrate-level meaning. The steady state is the substrate’s equilibrium distribution of alignment commitments under continuous drive + dissipation. Local fluctuations are damped; global topological structure (curvature, Chern number) is preserved.

2.7 P-PC-7. Global ED-curvature on the Brillouin zone

The *global ED-curvature* on the Brillouin zone is the rule-type curvature two-form $\Omega_n(\mathbf{k})$ defined on the closed parameter manifold $\mathcal{B} = \mathbb{T}^2$ (the 2D Brillouin zone is topologically a torus). It is gauge-invariant (§4.4 below) and integrates over $\mathcal{B}$ to a quantity that is integer-valued in units of 2π (Appendix A).

Substrate-level meaning. The global ED-curvature is the substrate-level measure of rule-type twist accumulated per unit area of the parameter manifold. It is the curvature of the rule-type connection on the BZ-bundle of band eigenchannels.

No additional substrate primitives are introduced beyond P-PC-1 through P-PC-7.

3 Constructing the Synthetic Frequency Lattice

3.1 Frequency modes as participation-rule indices

In the optical-fiber-loop experiment, the cavity supports a discrete set of resonant frequency modes $\omega_m = \omega_0 + m\Omega_R$ for $m \in \mathbb{Z}$. Each frequency mode is a distinct timing alignment of ED-flow. Electro-optic modulation at frequencies commensurate with Ω_R couples adjacent modes, and at frequencies commensurate with $2\Omega_R$ couples next-nearest modes.

The substrate-level identification: mode m is a participation rule r_m in the synthetic dimension (P-PC-2), and the chain's pre-individuation amplitude is

$$|\psi\rangle = \sum_m c_m |m\rangle, \quad \sum_m |c_m|^2 = 1.$$

3.2 NN and NNN couplings from electro-optic modulation

A modulation pulse at frequency Ω_R with complex amplitude $t_1 e^{i\theta_1}$ couples mode m to mode $m+1$. A modulation pulse at frequency $2\Omega_R$ with complex amplitude $t_2 e^{i\phi}$ couples mode m to mode $m+2$. The effective Hamiltonian on the synthetic lattice is

$$H_{\text{synth}} = -t_1 \sum_m (|m+1\rangle\langle m| + \text{h.c.}) - t_2 \sum_m (e^{i\phi} |m+2\rangle\langle m| + \text{h.c.}).$$

Without loss of generality, we choose the phase of the NN coupling to be zero ($\theta_1 = 0$); any non-zero θ_1 can be absorbed by a global gauge transformation $|m\rangle \rightarrow e^{im\theta_1} |m\rangle$.

3.3 Two-sublattice structure (honeycomb analog)

The Haldane model lives on a honeycomb lattice with two sublattices A and B. The synthetic-dimension realization assigns even-indexed modes to sublattice A and odd-indexed modes to sublattice B. The NN couplings connect $A \leftrightarrow B$ (intra-cell), the NNN couplings connect $A \leftrightarrow A$ and $B \leftrightarrow B$ (inter-cell of the same sublattice), and a possible on-site mass term M distinguishes A from B sublattice energies.

Group the modes into unit cells of two: $|A_n\rangle = |2n\rangle$ and $|B_n\rangle = |2n+1\rangle$ for $n \in \mathbb{Z}$. The lattice period is $a = 2$ in synthetic-index units.

3.4 Translation operator and commutation

Define the translation operator T_a by

$$T_a |A_n\rangle = |A_{n+1}\rangle, \quad T_a |B_n\rangle = |B_{n+1}\rangle.$$

T_a is unitary on the synthetic-dimension Hilbert space (its inverse is T_{-a} acting by the opposite shift). The Hamiltonian translates as

$$T_a H_{\text{synth}} T_a^{-1} = H_{\text{synth}}$$

because: (i) the NN couplings shift to structurally identical NN couplings on the translated cells; (ii) the NNN couplings shift similarly with the same complex phase $e^{i\phi}$; (iii) the on-site mass term M is sublattice-dependent but translation-invariant. Therefore $[H_{\text{synth}}, T_a] = 0$.

3.5 Bloch form

By the spectral theorem for commuting Hermitian operators, H_{synth} and T_a admit a simultaneous eigenbasis. We solve $T_a \psi = \lambda \psi$ with $|\lambda| = 1$ (unitarity), so $\lambda = e^{ika}$ for some $k \in \mathcal{B} = (-\pi/a, \pi/a]$. The eigenvalue equation gives the Bloch form on the two-sublattice synthetic lattice:

$$|\psi_k\rangle = \sum_n e^{ikna} (u_k^A |A_n\rangle + u_k^B |B_n\rangle).$$

The two-component spinor $\mathbf{u}_k = (u_k^A, u_k^B)^T$ is the periodic factor on the unit cell.

3.6 Two-dimensional generalization

The Chénier et al. experiment realizes a 2D Haldane model in synthetic dimension by using two distinct modulation patterns to encode two effective lattice directions. The synthetic Brillouin zone is then a torus $\mathbb{T}^2 = \mathcal{B}_x \times \mathcal{B}_y$ with each factor a circle. The Bloch form generalizes to

$$|\psi_{\mathbf{k}}\rangle = e^{i\mathbf{k} \cdot \mathbf{n}_a} (u_{\mathbf{k}}^A |A\rangle + u_{\mathbf{k}}^B |B\rangle), \quad \mathbf{k} = (k_x, k_y) \in \mathbb{T}^2,$$

where $\mathbf{n}_a$ is the integer-valued unit-cell index in 2D.

4 Deriving the Haldane-Type Model as ED-Curvature

4.1 Bloch Hamiltonian

After Bloch decomposition, the Hamiltonian is block-diagonalized over $\mathbf{k} \in \mathbb{T}^2$, with each block a 2×2 matrix on the A/B sublattice space:

$$H(\mathbf{k}) = h_x(\mathbf{k})\sigma_x + h_y(\mathbf{k})\sigma_y + h_z(\mathbf{k})\sigma_z + \varepsilon(\mathbf{k})I,$$

where $\sigma_{x,y,z}$ are Pauli matrices on the sublattice space and $\varepsilon(\mathbf{k})$ is the diagonal kinetic-plus-mass contribution. For the Haldane construction:

$$\begin{aligned} h_x(\mathbf{k}) + ih_y(\mathbf{k}) &= -t_1 \sum_{\delta} e^{i\mathbf{k} \cdot \delta} \\ h_z(\mathbf{k}) &= M - 2t_2 \sin \phi \sum_{\delta_{\text{NNN}}} \sin(\mathbf{k} \cdot \delta_{\text{NNN}}) \\ \varepsilon(\mathbf{k}) &= -2t_2 \cos \phi \sum_{\delta_{\text{NNN}}} \cos(\mathbf{k} \cdot \delta_{\text{NNN}}) \end{aligned}$$

The Pauli-vector $\mathbf{h}(\mathbf{k}) = (h_x, h_y, h_z)$ encodes the rule-type structure on the sublattice space at each $\mathbf{k}$.

4.2 Eigenvalues and eigenchannels

The eigenvalues of $H(\mathbf{k})$ are

$$E_{\pm}(\mathbf{k}) = \varepsilon(\mathbf{k}) \pm |\mathbf{h}(\mathbf{k})|, \quad |\mathbf{h}(\mathbf{k})| = \sqrt{h_x^2 + h_y^2 + h_z^2}.$$

Two bands, gapped wherever $|\mathbf{h}(\mathbf{k})| > 0$. The eigenchannels are spinors on the sublattice space:

$$u_{-}(\mathbf{k}) = (\sin(\theta/2), -e^{i\varphi_h} \cos(\theta/2))^T, \quad u_{+}(\mathbf{k}) = (\cos(\theta/2), e^{i\varphi_h} \sin(\theta/2))^T,$$

where $(\theta(\mathbf{k}), \varphi_h(\mathbf{k}))$ are the spherical coordinates of the unit vector $\hat{\mathbf{h}}(\mathbf{k}) = \mathbf{h}/|\mathbf{h}|$:

$$\hat{\mathbf{h}}(\mathbf{k}) = (\sin \theta \cos \varphi_h, \sin \theta \sin \varphi_h, \cos \theta).$$

The map $\mathbf{k} \mapsto \hat{\mathbf{h}}(\mathbf{k})$ is a continuous map from the Brillouin zone $\mathbb{T}^2$ to the unit sphere S^2 .

4.3 Berry connection re-derived inline

For the lower band, the Berry connection on the BZ is

$$\mathcal{A}_-(\mathbf{k}) \equiv i \langle u_-(\mathbf{k}) | \nabla_{\mathbf{k}} u_-(\mathbf{k}) \rangle.$$

Compute $\langle u_- | \partial_{k_i} u_- \rangle$ using the spinor form above. After working through the spherical-coordinate algebra:

$$\langle u_- | \partial_i u_- \rangle = i \cos^2(\theta/2) \partial_i \varphi_h.$$

Therefore

$$\mathcal{A}_-(\mathbf{k}) = i \cdot \langle u_- | \nabla_{\mathbf{k}} u_- \rangle = -\cos^2(\theta(\mathbf{k})/2) \nabla_{\mathbf{k}} \varphi_h(\mathbf{k}).$$

Equivalently, using $\cos^2(\theta/2) = (1 + \cos \theta)/2$:

$$\mathcal{A}_-(\mathbf{k}) = -\frac{1}{2}(1 + \hat{h}_z(\mathbf{k})) \nabla_{\mathbf{k}} \varphi_h(\mathbf{k}).$$

4.4 Berry curvature: the substrate-level map to the sphere

Compute the Berry curvature in 2D:

$$\Omega_-(\mathbf{k}) = \partial_{k_x} \mathcal{A}_-^y - \partial_{k_y} \mathcal{A}_-^x.$$

A standard manipulation (using the spherical-coordinate identity for the integrand) gives

$$\Omega_-(\mathbf{k}) = -\frac{1}{2} \hat{\mathbf{h}}(\mathbf{k}) \cdot (\partial_{k_x} \hat{\mathbf{h}} \times \partial_{k_y} \hat{\mathbf{h}}).$$

This is the **pullback of the unit-sphere area form** by the map $\mathbf{k} \mapsto \hat{\mathbf{h}}(\mathbf{k})$. The integrated Berry curvature over $\mathbb{T}^2$ is

$$\iint_{\mathbb{T}^2} \Omega_-(\mathbf{k}) d^2k = -\frac{1}{2} \iint_{\mathbb{T}^2} \hat{\mathbf{h}} \cdot (\partial_{k_x} \hat{\mathbf{h}} \times \partial_{k_y} \hat{\mathbf{h}}) d^2k = -\frac{1}{2} \cdot 4\pi \cdot (\text{degree of } \hat{\mathbf{h}}),$$

where the *degree* of $\hat{\mathbf{h}} : \mathbb{T}^2 \rightarrow S^2$ is the integer counting how many times the image wraps the sphere.

4.5 Gauge transformation

Under a gauge transformation $|u_-(\mathbf{k})\rangle \rightarrow e^{i\chi(\mathbf{k})} |u_-(\mathbf{k})\rangle$, the Berry connection transforms as

$$\mathcal{A}_-(\mathbf{k}) \rightarrow \mathcal{A}_-(\mathbf{k}) - \nabla_{\mathbf{k}} \chi(\mathbf{k}).$$

The Berry curvature is gauge-invariant by the equality of mixed partials.

4.6 Concentration of Berry curvature near Dirac points

The Berry curvature is large where $\hat{\mathbf{h}}(\mathbf{k})$ varies rapidly. In the Haldane construction, $\hat{\mathbf{h}}$ varies most rapidly near the *Dirac points* $\mathbf{K}, \mathbf{K}'$ in the BZ where $h_x(\mathbf{k}) = h_y(\mathbf{k}) = 0$ and $\hat{\mathbf{h}}$ is determined entirely by the small h_z component.

At the Dirac points, $h_z = M \mp 3\sqrt{3}t_2 \sin \phi$ takes opposite signs (for the two valleys $\mathbf{K}, \mathbf{K}'$) when $M < 3\sqrt{3}|t_2 \sin \phi|$. The unit vector $\hat{\mathbf{h}}$ then points to opposite poles of S^2 at the two Dirac points, and the map $\mathbf{k} \mapsto \hat{\mathbf{h}}(\mathbf{k})$ has degree ± 1 — wrapping the sphere once.

4.7 Topological phase transition

The relative magnitudes of M (sublattice mass) and $3\sqrt{3}t_2 \sin \phi$ (NNN-induced topological mass) determine the band topology:

- $|M| > 3\sqrt{3}|t_2 \sin \phi|$: trivial phase, $\hat{\mathbf{h}}(\mathbf{k})$ has zero degree, integrated curvature zero, $C_- = 0$.
- $|M| < 3\sqrt{3}|t_2 \sin \phi|$: topological phase, $\hat{\mathbf{h}}$ has degree ± 1 , integrated curvature $\pm 2\pi$, $C_- = \pm 1$.

The phase transition occurs at $|M| = 3\sqrt{3}|t_2 \sin \phi|$, where the band gap closes at one Dirac point and the topology of $\hat{\mathbf{h}}$ undergoes a discontinuous change.

5 Chern Number as Integrated Curvature

5.1 Definition

For each band n , the **Chern number** is

$$C_n \equiv \frac{1}{2\pi} \iint_{\mathbb{T}^2} \Omega_n(\mathbf{k}) d^2k.$$

In the Haldane lower band: C_- is 0 in the trivial phase and ± 1 in the topological phase.

5.2 Gauge invariance

Ω_n is gauge-invariant (§4.5), so the integral is gauge-invariant: C_n is independent of the choice of phase for $|u_n(\mathbf{k})\rangle$.

5.3 Surface independence

For two different choices of integration domain bounded by the same closed curve, the integrals differ by the Berry phase around the boundary, which is zero modulo 2π for a single-valued gauge. On a closed manifold like the torus, the integrated curvature is therefore invariant under continuous deformations.

5.4 Integer-quantization

The integrated curvature on a closed 2D manifold is integer-valued in units of 2π :

$$C_n \in \mathbb{Z}.$$

Full proof in Appendix A. The proof uses the parameter-space bundle structure on the torus, the obstruction to a globally smooth gauge choice, the winding number of transition functions on patch overlaps, and Stokes' theorem.

5.5 Substrate-level meaning of the Chern number

The Chern number C_n is the integer-valued global topological invariant of the band's rule-type bundle over the closed Brillouin zone. It counts the number of times the unit-vector field $\hat{\mathbf{h}}(\mathbf{k})$ wraps the sphere S^2 as $\mathbf{k}$ traverses the BZ, with sign determined by orientation. C_n is the global ED-twist accumulated by the band's rule-type connection around the closed parameter manifold.

6 ED-Tension and Quantized Hall Drift

6.1 Adding ED-tension to the Hamiltonian

ED-tension (synthetic electric field) adds a linear gradient to the on-site energies (P-PC-5):

$$H_{\text{total}} = H_{\text{synth}} + \mathbf{F} \cdot \hat{\mathbf{x}},$$

where $\hat{\mathbf{x}}$ is the position operator on the synthetic lattice and $\mathbf{F}$ is the ED-tension vector.

6.2 Semiclassical equations of motion

We derive the semiclassical equations of motion for a chain in a single band in the presence of ED-tension. Start from the Heisenberg equation of motion for the position operator restricted to band n . A standard wave-packet derivation gives:

$$\hbar \dot{\mathbf{k}} = \mathbf{F}, \quad \hbar \dot{x}_i = \partial E_n(\mathbf{k}) / \partial k_i - \epsilon_{ij} F_j \Omega_n(\mathbf{k}),$$

where the second term is the *anomalous velocity* — the component of the velocity perpendicular to the ED-tension, proportional to the Berry curvature.

Derivation of the anomalous velocity. Construct a wave packet centered at $\mathbf{k}_0$ in momentum and $\mathbf{x}_0$ in position. The expectation value of position is the wave-packet center plus a Berry-connection correction:

$$\langle x_i \rangle = x_{0,i} + \mathcal{A}_n^i(\mathbf{k}_0).$$

Differentiating with respect to time:

$$\begin{aligned} \langle \dot{x}_i \rangle &= (\partial x_{0,i} / \partial k_j) (dk_j / dt) + (\partial \mathcal{A}_n^i / \partial k_j) (dk_j / dt) \\ &= (1/\hbar) \partial E_n / \partial k_i + (F_j / \hbar) (\partial_j \mathcal{A}_n^i) \\ &= (1/\hbar) [\partial E_n / \partial k_i - F_j \Omega_n^{ji}], \end{aligned}$$

where in the last step we used $\partial_j \mathcal{A}_n^i - \partial_i \mathcal{A}_n^j = -\Omega_n^{ji}$.

In 2D with $\mathbf{F} = F \hat{\mathbf{x}}$ and Berry curvature $\Omega_n(\mathbf{k}) = \Omega_n^z(\mathbf{k})$:

$$\hbar \dot{x} = \partial E_n / \partial k_x \text{ (longitudinal),} \quad \hbar \dot{y} = \partial E_n / \partial k_y - F \cdot \Omega_n \text{ (transverse, anomalous).}$$

6.3 Quantized transverse drift

Consider a chain initialized at one synthetic-dimension site and subjected to ED-tension F in the x -direction. The semiclassical EOM gives $\hbar \dot{k}_x = F$, so k_x traverses the BZ in time $T = 2\pi\hbar/(Fa)$ (one Bloch oscillation period). The transverse displacement during one Bloch oscillation is

$$\Delta y = \int_0^T \dot{y} dt = (1/\hbar) \int_0^T [\partial E_n / \partial k_y - F \Omega_n] dt.$$

The first term is zero by periodicity of $E_n(\mathbf{k})$ in k_x . The second term, after changing variables from t to k_x , gives

$$\Delta y = - \int_{-\pi/a}^{\pi/a} \Omega_n(k_x, k_y) dk_x.$$

Integrated over all k_y values populated by the chain:

$$\langle \Delta y \rangle_{\text{band}} = -\frac{a}{2\pi} \iint_{\mathbb{T}^2} \Omega_n d^2k = -a \cdot C_n.$$

The transverse drift per Bloch oscillation period is $|\Delta y| = a \cdot |C_n|$. It is quantized in units of the lattice spacing, with quantization integer equal to the Chern number of the occupied band.

6.4 Quantization is independent of ED-tension strength

The drift per cycle is $a \cdot C_n$ regardless of F . A larger F produces faster Bloch oscillations (smaller T), but the drift per cycle is unchanged. The integrated current is $\Delta y/T = (Fa^2/(2\pi\hbar)) \cdot C_n$ — proportional to F and to C_n . This is the photonic analog of the integer quantum Hall conductance.

6.5 Substrate-level reading

- ED-tension forces $\mathbf{k}$ to traverse the BZ at uniform rate.
- The chain's pre-individuation amplitude follows the band eigenchannel $|u_n(\mathbf{k}(t))\rangle$ adiabatically.
- The Berry curvature on the BZ provides an anomalous-velocity term.
- Integrated over a full BZ traversal, the transverse displacement is the global rule-type curvature of the band — quantized in integer units of $a \cdot C_n$.
- **Light does not need charge.** It needs a periodic rule-type substrate with non-zero Chern number, plus an imposed ED-tension to make the global curvature manifest as transverse displacement.

7 Driven-Dissipative Steady State

7.1 Lindblad form derived inline

Real photonic systems involve continuous injection of substrate participation (drive) and continuous extraction (dissipation). The substrate-level state is a density operator ρ obeying a master equation derived from the system-bath coupling.

We derive the Lindblad form. Begin with the full system-plus-bath Hamiltonian $H_{\text{tot}} = H_{\text{sys}} + H_{\text{bath}} + H_{\text{int}}$. Trace out the bath, assume: (i) factorized initial state, (ii) weak system-bath coupling, (iii) Markovian limit, (iv) rotating-wave approximation. The resulting master equation is

$$\partial_t \rho = -i[H_{\text{sys}}, \rho] + \sum_{\alpha} \left[L_{\alpha} \rho L_{\alpha}^{\dagger} - \frac{1}{2} \{ L_{\alpha}^{\dagger} L_{\alpha}, \rho \} \right].$$

The Lindblad operators L_{α} encode the system-bath coupling: $L_{\alpha} = \sqrt{\gamma_{\alpha}} a_{\alpha}$ for loss to mode α at rate γ_{α} , and similar for drive terms.

7.2 Steady-state existence

For generic Lindblad operators (those making the dynamics ergodic on the system Hilbert space), the master equation has a unique steady state ρ_{ss} with $\partial_t \rho_{\text{ss}} = 0$. The steady state is reached on a timescale $\tau_{\text{ss}} \sim 1/\gamma$.

7.3 Lindblad effect on band populations

Lindblad-type dissipation does *not* preserve coherence within the band — local fluctuations in pre-individuation amplitudes are damped. However, it does preserve the *band population* under generic conditions: if the system is initialized in band n and the Lindblad operators do not directly couple to band $m \neq n$ (or do so with rate suppressed by the band gap), the steady-state population is concentrated in band n .

Substrate-level statement. Dissipation coarse-grains the chain's pre-individuation amplitude. The global rule-type curvature of the band is unchanged by this coarse-graining, because curvature is gauge-invariant and Lindblad-type local dissipation does not deform the gauge structure of the band's $|u_n(\mathbf{k})\rangle$.

7.4 Steady-state Hall drift

Under continuous drive + dissipation + ED-tension, the chain reaches a steady state in which:

- The band population is concentrated in a single band n .
- The momentum distribution is biased by the ED-tension.
- The center-of-mass position drifts at rate proportional to C_n .

The steady-state center-of-mass displacement per Bloch oscillation is identical to the closed-system result: $\langle \Delta y \rangle_{\text{steady}} = -a \cdot C_n$. **The dissipation does not alter the quantization.**

7.5 Why dissipation cannot deform C_n

The Chern number is an integer. Lindblad-type dissipation acts continuously on the density operator. A continuous deformation cannot change an integer-valued topological invariant unless it crosses a singularity (band gap closure). Generic Lindblad dynamics in a gapped band do not close the band gap; the Chern number is therefore preserved.

This is the substrate-level statement of *topological protection*: continuous deformation of the substrate's rule-type structure within a topological phase cannot change the integer-valued global rule-type invariant.

8 Substrate-Level Reading

Standard photonic-Chern object	Substrate-level meaning
Photon	Minimal ED-channel (P-PC-1)
Synthetic dimension (frequency space)	Re-indexing of participation rules (P-PC-2)
Frequency mode	Distinct timing alignment of ED-flow
NN coupling (real)	Local ED-flow continuity between adjacent rules
NNN coupling (complex phase)	Global ED-twist on the rule-type lattice
Time-reversal breaking	Directional ED-bias (P-PC-4)
Synthetic electric field	ED-tension on rule-type structure (P-PC-5)
Bloch eigenchannel $\psi_{\mathbf{k}}$	Chain translating with substrate periodicity modulo quasi-momentum phase
Brillouin zone	Compact parameter-space domain ($\mathbb{T}^2$ in 2D)
Band $E_n(\mathbf{k})$	Eigen-energy of substrate's rule-type structure on the BZ
Band gap	Rule-type incompatibility — no eigenchannel at gap energy
Berry connection $\mathcal{A}_n(\mathbf{k})$	Rule-type connection on band's BZ-bundle
Berry curvature $\Omega_n(\mathbf{k})$	Rule-type curvature in parameter space; gauge-invariant
Chern number C_n	Integer-valued global ED-curvature of band's BZ-bundle
Anomalous velocity $\mathbf{F} \times \Omega_n$	ED-flow's transverse response under imposed tension
Quantized Hall drift $a \cdot C_n$ per cycle	Inevitable transverse displacement of minimal channel under tension on curved BZ-bundle
Lindblad steady state	Driven-dissipative ED-flow equilibrium pinning to global manifold
Topological protection	Integer invariant unchanged under continuous deformation within a phase

8.1 The substrate-level meaning of the experiment

The Chénier et al. experiment is not a photonic imitation of the electronic quantum Hall effect. It is a direct demonstration that the substrate-level rule-type-curvature mechanism is independent of the platform: when a minimal ED-channel (light) is forced through a periodic rule-type substrate (synthetic frequency lattice) with non-zero global rule-type curvature (Haldane-type Chern number) under imposed ED-tension (modulation detuning), the channel exhibits a quantized transverse drift whose quantization is the integer-valued global ED-curvature.

The same mechanism applies to electrons in magnetic fields, to phonons in engineered substrates, to mechanical modes in lattice resonators, and to any minimal ED-channel in any substrate carrying non-zero global rule-type curvature. The substrate-level mechanism is platform-independent.

9 What's Forced, What's Inherited, What's Open

What is FORCED by the substrate ontology

- Synthetic frequency lattice as periodic rule-type substrate.
- Bloch form $|\psi_{\mathbf{k}}\rangle = e^{i\mathbf{k}\cdot\mathbf{n}_a}|\mathbf{u}_{\mathbf{k}}\rangle$.
- Two-band Bloch Hamiltonian with Pauli structure on the sublattice space.
- Berry connection $\mathcal{A}_n(\mathbf{k}) = i\langle u_n|\nabla_{\mathbf{k}}u_n\rangle$.
- Gauge-invariance of Berry curvature.
- Berry curvature as pullback of the unit-sphere area form under $\mathbf{k} \mapsto \hat{\mathbf{h}}(\mathbf{k})$.
- Topological phase transition at $|M| = 3\sqrt{3}|t_2 \sin \phi|$.
- Chern-number integer-quantization $C_n \in \mathbb{Z}$ (Appendix A).
- Quantized Hall drift $\Delta y = -a \cdot C_n$ per Bloch oscillation.
- Topological protection under driven-dissipative dynamics.

What is FORM-FORCED-INHERITED (and re-derived inside this document)

Spectral theorem for commuting Hermitian operators; unitarity of translation operator; Pauli-matrix algebra; spherical-coordinate parameterization; Lindblad master equation; wave-packet semiclassical EOM; anomalous-velocity term; Stokes' theorem on patches of the torus.

What remains OPEN

Higher-Chern manifolds and 4D quantum Hall; non-Hermitian topology; non-Abelian Berry curvature for degenerate bands; Floquet topological insulators; bulk-boundary correspondence and edge-state count; many-body topology (fractional quantum Hall, fractional Chern insulators).

10 What This Argument Establishes

This walkthrough establishes the following exact claims:

Claim 1. A periodic rule-type structure in synthetic dimension FORCES a discrete translation symmetry, a Bloch-form decomposition of eigenchannels, and a Brillouin-zone parameter manifold. In two dimensions, the BZ is topologically a torus $\mathbb{T}^2$.

Claim 2. Complex NNN couplings with phase $\phi \neq 0, \pi$ break time-reversal symmetry and induce a non-zero Berry curvature on the BZ. The Berry curvature is the pullback of the unit-sphere area form under the map $\mathbf{k} \mapsto \hat{\mathbf{h}}(\mathbf{k})$.

Claim 3. The Chern number $C_n = (1/2\pi) \iint_{\mathbb{T}^2} \Omega_n d^2k$ is integer-valued (Appendix A). The Haldane phase transition between $|M| < 3\sqrt{3}|t_2 \sin \phi|$ (topological, $C_- = \pm 1$) and $|M| > 3\sqrt{3}|t_2 \sin \phi|$ (trivial, $C_- = 0$) is FORCED by the band-gap closure.

Claim 4. Under imposed ED-tension $\mathbf{F}$, a chain in band n exhibits quantized transverse displacement $\Delta y = -a \cdot C_n$ per Bloch oscillation period. The quantization is FORCED by the BZ being closed and by the integer-valued Chern number.

Claim 5. Driven-dissipative dynamics described by Lindblad master equations preserve the band's Chern number under continuous deformation within a topological phase. The quantized Hall drift is robust against dissipation.

Claim 6 (negative). No new substrate primitives are required. The photonic Chern insulator and quantized Hall drift decompose into composition of P-PC-1 through P-PC-7, plus standard linear algebra, differential geometry, and the Chern integer-quantization theorem (Appendix A) re-derived as needed.

Claim 7 (scope-limit). This walkthrough does not derive: higher-Chern manifolds and 4D quantum Hall; non-Hermitian topology; non-Abelian Berry curvature; Floquet topological insulators; bulk-boundary correspondence; many-body topology.

The unified statement. Quantized Hall drift of light is the substrate-level inevitable transverse response of a minimal ED-channel propagating through a periodic rule-type substrate with non-zero global ED-curvature, under imposed ED-tension. Light does not need charge. It needs a curved rule-type manifold and a tension to traverse it.

A Appendix A: Chern-Number Integer Quantization

This appendix provides a self-contained proof that

$$C_n = \frac{1}{2\pi} \iint_{\mathbb{T}^2} \Omega_n(\mathbf{k}) d^2k \in \mathbb{Z}$$

for any smooth Berry curvature Ω_n arising from a smooth rank-1 eigenchannel section of a $U(1)$ bundle over the torus $\mathbb{T}^2$.

A.1 Setup: the $U(1)$ bundle structure

Fix band n . The eigenchannel $|u_n(\mathbf{k})\rangle$ is smooth in $\mathbf{k}$ over any open patch on which a smooth gauge can be chosen. The space of all eigenchannels at all $\mathbf{k}$ with $|u_n\rangle$ defined up to a $U(1)$ phase forms a $U(1)$ principal bundle $P \rightarrow \mathbb{T}^2$ over the torus.

Topological obstruction. When $C_n \neq 0$, no globally smooth section of P exists — the bundle is non-trivial. To define $|u_n(\mathbf{k})\rangle$ everywhere, the BZ must be covered by overlapping patches with different gauge choices, and the relations between gauge choices on overlaps are encoded in *transition functions*.

A.2 Two-patch covering of the torus

Cover $\mathbb{T}^2$ by two patches U_1 and U_2 such that their intersection $U_{12} = U_1 \cap U_2$ is a small annular neighborhood of a non-contractible loop γ on the torus.

Choose smooth gauges $|u_n^{(1)}(\mathbf{k})\rangle$ on U_1 and $|u_n^{(2)}(\mathbf{k})\rangle$ on U_2 . On the overlap U_{12} , the two gauge choices differ by a $U(1)$ -valued transition function:

$$|u_n^{(2)}(\mathbf{k})\rangle = e^{i\chi_{12}(\mathbf{k})} |u_n^{(1)}(\mathbf{k})\rangle \quad \text{for } \mathbf{k} \in U_{12},$$

where $\chi_{12} : U_{12} \rightarrow \mathbb{R}/2\pi\mathbb{Z}$ is the transition function (defined modulo 2π).

A.3 Berry connection on each patch

By the gauge transformation argument, the Berry connections on the two patches differ by

$$\mathcal{A}_n^{(2)}(\mathbf{k}) = \mathcal{A}_n^{(1)}(\mathbf{k}) - \nabla_{\mathbf{k}} \chi_{12}(\mathbf{k}) \quad \text{on } U_{12}.$$

The Berry curvature $\Omega_n = \partial_x \mathcal{A}_n^y - \partial_y \mathcal{A}_n^x$ is gauge-invariant and is therefore a globally well-defined smooth two-form on $\mathbb{T}^2$.

A.4 Stokes' theorem on each patch

Apply Stokes' theorem on each patch separately. Choose U_1 to be a topological disk (simply-connected) and U_2 to be a topological annulus around a non-contractible loop. By Stokes:

$$\iint_{U_1} \Omega_n d^2k = \oint_{\partial U_1} \mathcal{A}_n^{(1)} \cdot d\mathbf{k}.$$

For a torus covered by two patches, the total integrated curvature decomposes as

$$\iint_{\mathbb{T}^2} \Omega_n = \iint_{U_1} \Omega_n + \iint_{U_2} \Omega_n.$$

A.5 The transition-function winding

On the overlap U_{12} , the difference between the two gauge connections is $-\nabla\chi_{12}$. The line integral around the shared boundary gives:

$$\oint_{\partial U_1} (\mathcal{A}_n^{(2)} - \mathcal{A}_n^{(1)}) \cdot d\mathbf{k} = -[\chi_{12}(\text{end}) - \chi_{12}(\text{start})].$$

For a path ∂U_1 that closes, the difference $\chi_{12}(\text{end}) - \chi_{12}(\text{start})$ is the *winding number* of χ_{12} around the loop, multiplied by 2π :

$$\chi_{12}(\text{end}) - \chi_{12}(\text{start}) = 2\pi m, \quad m \in \mathbb{Z},$$

where m is the integer winding number. This is the only source of integer-valued contributions: χ_{12} is itself defined only modulo 2π .

A.6 Integration of the transition-function winding

Apply Stokes' theorem on each patch and add. The boundary ∂U_2 is traversed in the opposite direction. On the shared boundary:

$$\begin{aligned} \iint_{\mathbb{T}^2} \Omega_n &= \oint_{\partial U_1} \mathcal{A}_n^{(1)} \cdot d\mathbf{k} - \oint_{\partial U_1} \mathcal{A}_n^{(2)} \cdot d\mathbf{k} \\ &= \oint_{\partial U_1} (\mathcal{A}_n^{(1)} - \mathcal{A}_n^{(2)}) \cdot d\mathbf{k} = \oint_{\partial U_1} \nabla\chi_{12} \cdot d\mathbf{k} = 2\pi m, \quad m \in \mathbb{Z}. \end{aligned}$$

Therefore:

$$C_n = \frac{1}{2\pi} \iint_{\mathbb{T}^2} \Omega_n = m \in \mathbb{Z}.$$

A.7 Substrate-level reading of the proof

The integer-quantization arises from the substrate-level fact that the rule-type connection on the BZ-bundle is *single-valued modulo 2π as a $U(1)$ -valued connection*. The transition function χ_{12} between two gauge choices must be $U(1)$ -valued — single-valued on the overlap. Around a closed loop, the only single-valued $U(1)$ -valued differences are integer multiples of 2π .

The Chern number counts the total winding of these transition functions across the BZ. It is the substrate-level statement of *how non-trivially the rule-type bundle is glued together* across the closed parameter manifold. A trivial bundle has zero winding ($C_n = 0$); a non-trivial bundle has integer-valued winding corresponding to the Chern number.

A.8 Why the proof requires the BZ to be closed

If the BZ were an open manifold, no closed-loop integral would arise, and the transition-function winding argument would not give a well-defined integer. The integer-quantization is intrinsically a property of *closed* parameter manifolds. The substrate-level requirement that the BZ be closed (forced by the periodicity of the substrate’s rule-type structure) is the load-bearing condition for integer-quantization.

A.9 Higher Chern numbers

For higher-dimensional closed manifolds, additional Chern numbers arise (second Chern number for $\mathbb{T}^4$, etc.). The proof structure generalizes: the obstruction to a globally smooth gauge is captured by transition-function windings on the manifold’s higher-dimensional cycles. For the present walkthrough’s scope (2D photonic Chern insulator), the first Chern number C_n derived above is sufficient.

References

- Haldane, F. D. M. *Model for a quantum Hall effect without Landau levels*. Phys. Rev. Lett. **61**, 2015 (1988).
- Thouless, D. J., Kohmoto, M., Nightingale, M. P., den Nijs, M. *Quantized Hall conductance in a two-dimensional periodic potential*. Phys. Rev. Lett. **49**, 405 (1982).
- Berry, M. V. *Quantal phase factors accompanying adiabatic changes*. Proc. R. Soc. A **392**, 45 (1984).
- Xiao, D., Chang, M.-C., Niu, Q. *Berry phase effects on electronic properties*. Rev. Mod. Phys. **82**, 1959 (2010).
- Lindblad, G. *On the generators of quantum dynamical semigroups*. Comm. Math. Phys. **48**, 119 (1976).
- Gorini, V., Kossakowski, A., Sudarshan, E. C. G. *Completely positive dynamical semigroups of N -level systems*. J. Math. Phys. **17**, 821 (1976).
- Ozawa, T., Carusotto, I. *Topological photonics*. Nat. Photon. **8**, 821 (2014).
- Lustig, E., Segev, M. *Topological photonics in synthetic dimensions*. Adv. Opt. Photonics **13**, 426 (2021).
- Chénier, A., d’Aligny, B., Pellerin, F., Blanchard, P., Ozawa, T., Carusotto, I., St-Jean, P. *Quantized Hall Drift in a Frequency-Encoded Photonic Chern Insulator*. Phys. Rev. X **16**, 1 (2026).
- Nakahara, M. *Geometry, Topology, and Physics*. IOP Publishing (2003).